Ayush Jain

Research Scientist, Meta PhD, Computer Science, USC

Reinforcement Learning, LLM Post-Training and Agents

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Reinforcement learning researcher with production LLM post-training experience at Meta and prior work training and evaluating VLM agents for Android computer use and long-horizon instruction following in Minecraft.

Professional Experience

Meta Feb 2025 – Present
Research Scientist, Applied Reinforcement Learning Team, Monetization Sunnyvale, CA

- **Production LLM post-training.** Post-trained LLM conversation agents with **SFT and GRPO** for Messenger Business Messaging; developed multi-turn LLM-judge evaluation.
- **LLM reasoning model for generative ad ranking.** Developing a model that reasons about ad quality for individual users and listwise relationships among candidate ads.
- **Multi-step RL for long-term ad relevance.** Building Meta's first multi-step RL system for ad retrieval and ranking to improve long-term ad quality, align with user interests, build trust, and encourage purchases. Lead modeling with CQL, IQL, and TD3-BC; built large-scale offline RL training and evaluation infrastructure.

Meta Reality Labs Research May 2024 – Dec 2024
Research Intern with Nitin Kamra Redmond, WA

- **LLM judge for Android computer-use agents.** Developed an evaluation and training pipeline for VLM agents, comparing parsed screenshots across trajectories to assess progress toward user instructions. Used internally for training data curation and extended to human labeling.
- Trained VLM computer-use policies with **advantage-weighted regression (AWR)** for natural-language instruction following.

Microsoft Research May 2023 – Jan 2024
Research Intern with Eric Yuan and Marc-Alexandre Côté Montreal, Canada

- **Long-horizon VLM agents.** Trained a VLM agent for long-horizon instruction following by distilling trajectories from an automated data-collection policy driven by a Voyager-style Minecraft agent with privileged information.

Naver AI Research 2021, 2022
Research Intern with Kyung-Min Kim and Joseph J. Lim Seoul, South Korea (remote)

- **RL in complex action spaces.** Researched RL with large, varying, and slate action spaces for physical reasoning and recommender systems.
- **Action-aware policies.** Built graph attention policies over available items and their relationships, improving simulated slate recommendation and supporting reasoning over sets of physical tools.

Samsung Research Sep 2016 – Jun 2018
Research Engineer, Data Analytics Lab with James Geraci and Yunsu Lee Seoul, South Korea

- Demonstrated that LSTM time-series models outperformed ARIMA for Samsung's market-share prediction tasks; also worked on time-series anomaly detection.

Reinforcement Learning Foundations

PhD Research, University of Southern California 2018 – 2024
Thesis: Decision-Making in Complex Action Spaces

- **Policy optimization and multi-task behavior sharing.** Developed methods to escape local optima in complex Q-functions and selectively share behavior across task-specific policies.
- **Generalization in complex action spaces.** Developed RL methods for unseen and varying action sets, including learned action representations and graph attention policies; designed the CREATE environment.

RLC 2025 Outstanding Paper Award on Empirical Reinforcement Learning Research

Publications and Patent

Conference Papers

[1] **Credit Assignment with Resets in Language Model Reasoning**. **NeurIPS 2026**.
A. Samanta, A. Magesh, **A. Jain**, Y. Yu, D. Jiang, K. Asadi, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni.

[2] **Structure Enables Effective Self-Localization of Errors in LLMs**. **ICML 2026**.
A. Samanta, A. Magesh, **A. Jain**, K. Asadi, Y. Yu, D. Jiang, B. Vidolov, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni.

[3] **When a Robot is More Capable than a Human: Learning from Constrained Demonstrators**. **ICLR 2026**.
X. Li, **A. Jain**, Z. Yang, Y. Korkmaz, E. Bıyık.

[4] **Imbalanced Gradients in RL Post-Training of Multi-Task LLMs**. **EACL 2026**.
R. Wu, A. Samanta, **A. Jain**, S. Fujimoto, J. Kwon, B. Kretzu, Y. Yu, K. Hassani, B. Vidolov, Y. Efroni.

[5] **Actor-Free Continuous Control via Structurally Maximizable Q-Functions**. **NeurIPS 2025**.
Y. Korkmaz*, U. Bhuwania*, **A. Jain**†, E. Bıyık†.

[6] **Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions**. **RLC 2025**.
A. Jain, N. Kosaka, X. Li, K. Kim, E. Bıyık, J. Lim.
Outstanding Paper Award on Empirical Reinforcement Learning Research

[7] **QMP: Q-switch Mixture of Policies for Multi-Task Behavior Sharing**. **ICLR 2025**.
G. Zhang*, **A. Jain***, I. Hwang, S. Sun, J. Lim.

[8] **Know Your Action Set: Learning Action Relations for Reinforcement Learning**. **ICLR 2022**.
A. Jain*, N. Kosaka*, K. Kim, J. Lim.

[9] **Generalization to New Actions in Reinforcement Learning**. **ICML 2020**.
A. Jain*, A. Szot*, J. Lim.

Workshop Papers and Presentations

[10] **Anticipating Unintended Robot Behaviors with Side Effect Critics**. **RSS 2026 Workshop**.
R. Lindeborg, R. J. Peng, **A. Jain**, J. Zhang, A. Anwar, J. Thomason.
Rethinking What It Means to be "Safe" for Generalist Robots.
Spotlight presentation

[11] **Uniform Information Density Effects on Syntactic Choice in Hindi**. **COLING 2018 Workshop**.
A. Jain, V. Singh, S. Ranjan, R. Rajkumar, S. Agarwal. *Workshop on Linguistic Complexity and NLP.*

[12] **Uniform Information Density models for language production**. **DGfS 2017**.
A. Jain, V. Singh, S. Agarwal, R. Rajkumar. *39th Annual Conference of the German Linguistic Society.*

Preprints

[13] **Training Small Language Models for Transferable API Selection**. **arXiv preprint, 2026**.
Z. Zhou, A. Magesh, J. Kwon, A. Kuriksha, D. Jiang, A. Zanette, **A. Jain**.

[14] **BayesBench: Evaluating LLM Belief Trajectories Under Multi-Turn Evidence Accumulation**. **arXiv preprint, 2026**.
A. Samanta, A. Magesh, T. Lancewicki, **A. Jain**, Y. Yu, P. Sajda, K. Hassani, A. Modi, D. Jiang, Y. Efroni.

[15] **Self-Improvement of Language Models by Post-Training on Multi-Agent Debate**. **arXiv preprint, 2025**.
A. Samanta, A. Magesh, R. Wu, **A. Jain**, Y. Yu, D. Jiang, B. Vidolov, P. Sajda, Y. Efroni, K. Hassani.

Patent

[16] **Method and Apparatus for Managing Operation Data of Appliance for Failure Prediction**. **U.S. Patent No. 11,182,235, Nov 23, 2021**.
J. Lee, M. Kim, **A. Jain**, T. Hwang, J. Kim, H. Cho.

* Equal contribution. † Equal advising.

Education

University of Southern California Dec 2024
PhD and MS in Computer Science GPA: 4.0/4.0
Advisors: Joseph J. Lim and Erdem Bıyık. **Thesis: Decision-Making in Complex Action Spaces**

Indian Institute of Technology Delhi Jun 2016
B.Tech in Electrical Engineering

RL Environment and Benchmark Design

- **CREATE:** Developed an environment for sequential decision-making and generalization to unseen actions.
- **Restricted MuJoCo:** Developed continuous-control tasks exposing policy-gradient failure modes in complex Q-functions.
- **Multistage Reacher (QMP):** Developed a simultaneous multi-task RL environment with shared and conflicting subgoals for evaluating selective behavior sharing.

Research Mentoring

Norio Kosaka (Naver AI → PhD student at NII Japan) ICLR 2022, RLC 2025
Xinhu Li (USC → PhD student at NUS) RLC 2025, ICLR 2026
Urvi Bhuwania (USC) NeurIPS 2025
Zhaoyi Zhou (CMU, Meta research intern) Current

Teaching and Service

Teaching Assistant, University of Southern California
Deep Learning and its Applications, CSCI-566.
Spring and Fall 2019; Fall 2020; Spring 2023; Spring and Fall 2024.

Reviewer

- ICLR 2023–2027
- NeurIPS 2023–2026
- NeurIPS Workshop Proposals 2026
- ICML 2025–2026
- CoRL 2021–2024
- RLC 2025–2026
- AAAI 2026–2027
- ACL 2027

Selected Honors

- **Outstanding Paper Award on Empirical Reinforcement Learning Research**, RLC 2025.
- **Silver Reviewer Award**, ICML 2026.
- **Travel Grant**, RLC 2025.